

Question Bank



**MALLA REDDY
UNIVERSITY**

(Telangana State Private Universities Act No.13 of 2020 and
G.O.Ms.No.14, Higher Education (UE) Department)

Operating Systems

II B.Tech I Semester

(2025-26)

Department of CSE (Data Science)

School of Engineering



School of Engineering-A.Y 2025-2026 Regulation: R24
(CSE-DATA SCIENCE)

II Year B. Tech - I Semester
Subject Name: OPERATING SYSTEMS
Subject Code: MR24-1CS0301

QUESTION BANK

Qno	Question	Marks	Section												
1	Define Operating systems. Explain about various types of operating Systems?	8	Section-I												
2	Explain the basic system components of a computer system with a neat diagram	8	Section-I												
3	a) Compare Simple Batch and Real-Time Operating Systems b) illustrating their architecture and evaluating their advantages and disadvantages	4 4	Section-I												
4	Describe the key services offered by an operating system in detail?	8	Section-I												
5	Analyze the core functionalities of an operating system.	8	Section-I												
6	Explain the different structuring methods used in the OS with neat sketch?	8	Section-I												
7	Discuss the different Design Issues related to the Operating System.	8	Section-I												
8	Define system calls. Classify the types of system calls and illustrate each with suitable examples.	8	Section-I												
9	Explain the characteristics of Parallel, Distributed, and Real-Time Operating Systems.	8	Section-I												
10	a) Differentiate between user mode and kernel mode. b) Describe how a system call switches the mode of execution.	4 4	Section-I												
11	Define process. Describe the different process states and illustrate them using a labeled diagram.	8	Section-II												
12	Define Threads? Classify the different types of threads	8	Section-II												
13	Analyze the role and behavior of cooperating processes in operating systems.	8	Section-II												
14	Demonstrate the key operations performed on a process, such as creation, termination, and scheduling.	8	Section-II												
15	Evaluate different multi-threading models and their impact on performance and resource management.	8	Section-II												
16	Compare Preemptive and Non-Preemptive Scheduling with suitable examples.	8	Section-II												
17	Explain about the scheduling criteria for CPU scheduling with examples.	8	Section-II												
18	a) Describe the First-Come, First-Served (FCFS) scheduling algorithm b) Calculate the Average waiting time and average Turnaround time. Provide the Gantt chart for the FCFS Scheduling Algorithm .	4 4	Section-II												
<table><tr><td>Process</td><td>P0</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td></tr><tr><td>Arrival Time(AT)</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td></tr></table>		Process	P0	P1	P2	P3	P4	Arrival Time(AT)	0	2	4	6	8		
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19	a) Explain about the SJF Scheduling Algorithm. b) Calculate the Average waiting time and average Turnaround time. Provide the Gantt chart for the SJF Scheduling Algorithm. <table><tr><td>Process</td><td>P0</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td></tr><tr><td>Arrival Time(AT)</td><td>0</td><td>1</td><td>3</td><td>5</td><td>9</td></tr><tr><td>Burst Time (BT)</td><td>2</td><td>3</td><td>4</td><td>6</td><td>1</td></tr></table>	Process	P0	P1	P2	P3	P4	Arrival Time(AT)	0	1	3	5	9	Burst Time (BT)	2	3	4	6	1	4 4	Section-II																																																															
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20	Find the Average waiting time and Turnaround Time for Round Robin Scheduling Algorithm (Time Quantum=2 milliseconds). <table><tr><td>Process</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td><td>P6</td></tr><tr><td>Arrival Time(AT)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>6</td></tr><tr><td>Burst Time (BT)</td><td>4</td><td>5</td><td>2</td><td>1</td><td>6</td><td>3</td></tr></table>	Process	P1	P2	P3	P4	P5	P6	Arrival Time(AT)	0	1	2	3	4	6	Burst Time (BT)	4	5	2	1	6	3	8	Section-II																																																												
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21	Define Deadlock. Identify the necessary conditions for deadlock and describe each characteristic.	8	Section-III																																																																																	
22	Discuss in detail the techniques of deadlock prevention.	8	Section-III																																																																																	
23	a) What is the purpose of Banker's Algorithm in operating systems? b) Consider the following table of a system. Using the Banker's Algorithms. Resource type R1 has 10 instances, Resource type R2 has 5 instances and R3 has 7 instances. <table><tr><th rowspan="2">Process</th><th colspan="3">Allocation</th><th colspan="3">Maximum</th><th colspan="3">Available</th><th colspan="3">Need</th></tr><tr><th>R1</th><th>R2</th><th>R3</th><th>R1</th><th>R2</th><th>R3</th><th>R1</th><th>R2</th><th>R3</th><th>R1</th><th>R2</th><th>R3</th></tr><tr><td>P0</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td><td>7</td><td>4</td><td>3</td></tr><tr><td>P1</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td colspan="3" rowspan="4"></td><td>1</td><td>2</td><td>2</td></tr><tr><td>P2</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td>6</td><td>0</td><td>0</td></tr><tr><td>P3</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td>0</td><td>1</td><td>1</td></tr><tr><td>P4</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td>4</td><td>3</td><td>1</td></tr></table>	Process	Allocation			Maximum			Available			Need			R1	R2	R3	R1	R2	R3	R1	R2	R3	R1	R2	R3	P0	0	1	0	7	5	3	3	3	2	7	4	3	P1	2	0	0	3	2	2				1	2	2	P2	3	0	2	9	0	2	6	0	0	P3	2	1	1	2	2	2	0	1	1	P4	0	0	2	4	3	3	4	3	1	4 4	Section-III
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P4	0	0	2	4	3	3				4	3	1																																																																								
24	Describe the deadlock detection algorithm. How is recovery from deadlock carried out?	8	Section-III																																																																																	
25	Explain the Semaphores and list out its advantages and disadvantages?	8	Section-III																																																																																	
26	Analyze any two classical synchronization problems and their significance in process coordination.	8	Section-III																																																																																	
27	a) Describe the Dining Philosophers problem. b) Justify how a deadlock can occur in the classic solution.	4 4	Section-III																																																																																	
28	a) Implement a semaphore-based solution for the bounded-buffer problem b) justify the roles of mutex, empty, and full semaphores.	4 4	Section-III																																																																																	
29	Compare and contrast semaphores and monitors, highlighting their strengths and weaknesses.	8	Section-III																																																																																	
30	Explain how synchronization hardware, such as the TestAndSet() instruction, can be used to solve the critical-section problem.	8	Section-III																																																																																	
31	Compare pipes and message queues, including their suitability for different IPC scenarios.	8	Section-IV																																																																																	

32	Explain shared memory IPC, its synchronization challenges, and how semaphores address them.	8	Section-IV
33	Differentiate between ordinary pipes and FIFOs, noting their usage for related versus unrelated processes.	8	Section-IV
34	Explain the message-passing model, including blocking and non-blocking communication primitives.	8	Section-IV
35	Analyze the concept of segmentation with paging and how it enhances memory management efficiency.	8	Section-IV
36	Explain about Virtual Memory and Demand Paging?	8	Section-IV
37	Compute the number of page faults for FIFO page replacement strategy for the given reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 with three page frames.	8	Section-IV
38	Distinguish between logical and physical address space in memory management.	8	Section-IV
39	Illustrate the concept of paging in memory management using a labeled diagram	8	Section-IV
40	Compare two page replacement algorithms, such as FIFO and LRU, using a simple example.	8	Section-IV
41	List and briefly define the three main file access methods: sequential, direct, and indexed sequential	8	Section-V
42	What is a Directory? Classify the different directory structures used in File system.	8	Section-V
43	Explain in detail about Allocation Methods in File.	8	Section-V
44	Discuss about the File Structure?	8	Section-V
45	Identify common file I/O system calls and demonstrate their usage with examples.	8	Section-V
46	Discuss about the Disk Management System in OS?	8	Section-V
47	a) What are FCFS and SSTF disk scheduling algorithms? b) Calculate the FCFS and SSTF disk scheduling algorithm with given Values?The request tracks in the order received are 98,183,37,122,14,124,65, 67. Apply the Following disk scheduling algorithms. Starting track at 53.	4 4	Section-V
48	a) What are SCAN and C-SCAN disk scheduling algorithm? b) Calculate the SCAN and C-SCAN disk scheduling algorithm with Given Values?The request tracks in the order received are 98,183,37,122,14,124,65, 67. Apply the Following disk scheduling algorithms. Starting track at 53.	4 4	Section-V
49	Explain the different system calls in file I/O a) Read & Write b) Close & Open	4 4	Section-V
50	Examine how file protection mechanisms prevent unauthorized access and ensure data security in multi-user environments.	8	Section-V